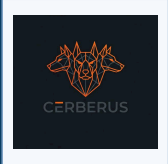


US Growth Nowcast & Surprise Index



INTERNAL

Macro surprise index from BEA, BLS, and FRED releases tested against SPY forward returns.

Author: Cerberus Research

Generated: 2026-08-07 08:47 UTC

N	EDGE	HIT-RATE
1	-3.86%	0.0%
CLASSIFICATION	internal	
AUTHOR	Cerberus Research	
RUN ID	20260807_094720	
GIT COMMIT	978cf61e	
DURATION	62.2s	
DATASETS	bea/bea_curated, bls/bls_curated, fred_latest/fred_latest_curated	
PARAMETER · LOOKBACK_YEARS	15	
PARAMETER · HORIZON	21	
PARAMETER · GROWTH_SERIES	{ 'bea/bea_curated': ['gdp_real', 'exports_goods_services', 'personal_consumption_expenditures', 'private_fixed_investment_equipment'], 'bls/bls_curated': ['nonfarm_payrolls', 'avg_hourly_earnings_total_private'], 'fred_latest/fred_latest_curated': ['industrial_production', 'payrolls', 'real_gdp'] }	
KPI	Value	
N	1	

EDGE
-3.86%

HIT-RATE
0.0%

KEY FINDINGS

- Latest growth surprise index: 1.918
- bea/bea_curated:bea_gdp_real: latest YoY change +44.85%
- bea/bea_curated:bea_exports_goods_services: latest YoY change +344.05%

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Data Sources

- **bea/bea_curated**: BEA national accounts series
- **bls/bls_curated**: BLS employment and price series
- **fred_latest/fred_latest_curated**: FRED macro series

Key Findings

- Latest growth surprise index: 1.918
- bea/bea_curated:bea_gdp_real: latest YoY change +44.85%
- bea/bea_curated:bea_exports_goods_services: latest YoY change +344.05%
- bea/bea_curated:bea_personal_consumption_expenditures: latest YoY change +1248.27%
- bea/bea_curated:bea_private_fixed_investment_equipment: latest YoY change +18829.25%
- fred_latest/fred_latest_curated:fred_latest_industrial_production: latest YoY change +120.27%
- fred_latest/fred_latest_curated:fred_latest_payrolls: latest YoY change +207.80%
- fred_latest/fred_latest_curated:fred_latest_real_gdp: latest YoY change +498.69%

Executive Summary

NOTE

Builds a US growth surprise index from BEA, BLS, and FRED releases. Each series is converted to a z-score of its latest change vs trailing trend; the mean z-score forms the surprise index, which is tested against SPY forward returns.

Results & Interpretation

- Across 1 event, the forward return averaged -3.86%. Returns were positive 0.0% of the time.
- Across 1 event, the forward return averaged +0.60%. Returns were positive 100.0% of the time.
- Across 1 event, the forward return averaged +1.70%. Returns were positive 100.0% of the time.

Objectives

NOTE

Evaluate whether the studied signal produces a measurable, repeatable edge.

Quantify the reliability of the observed patterns across the sample.

Methodology

NOTE

Series are normalized to z-scores using trailing 36-month mean and standard deviation. Surprise index is the cross-sectional average z-score. Regimes are quintiles of the surprise index.

Charts



Figure 1. Growth surprise index
Growth surprise index

Statistical Results

surprise_index
2.54
2.92
0.27
2.30
2.07
1.97
0.15
0.18
-0.13
1.92
1.78
1.62
1.48
1.51
1.42
1.00
1.33
1.92

Table 1. US growth surprise index

series	latest_val...	latest_yoy_pct
bea/bea_curated:bea_gdp_real	1,900,983	+ 44.85%
bea/bea_curated:bea_exports_goods_services	3,217,418	+ 344.05%
bea/bea_curated:bea_personal_consumption_expenditures	726,464	+ 1,248.27%
bea/bea_curated:bea_private_fixed_investment_equipment	243,998	+ 18,829.25%
fred_latest/fred_latest_curated:fred_latest_industrial_production	100.24	+ 120.27%
fred_latest/fred_latest_curated:fred_latest_payrolls	147,314	+ 207.80%
fred_latest/fred_latest_curated:fred_latest_real_gdp	18,666.62	+ 498.69%

Table 2. Latest series YoY changes

regime	count	avg_fwd_ret_...	median_fwd_ret_...	hit_rate_pct
Q2	1	-3.86%	-3.86%	0.00%
Q3	1	+ 0.60%	+ 0.60%	+ 100.00%
Q4	1	+ 1.70%	+ 1.70%	+ 100.00%

Table 3. Forward returns by surprise quintile

Limitations

NOTE

Results are research-grade, not investment advice. The sample is limited to available data, and measured edges may not persist out-of-sample.

Appendix Tables

latest_changes

series	latest_val...	latest_yoy_pct
bea/bea_curated:bea_gdp_real	1,900,983	+ 44.85%
bea/bea_curated:bea_exports_goods_services	3,217,418	+ 344.05%
bea/bea_curated:bea_personal_consumption_expenditures	726,464	+ 1,248.27%
bea/bea_curated:bea_private_fixed_investment_equipment	243,998	+ 18,829.25%
fred_latest/fred_latest_curated:fred_latest_industrial_production	100.24	+ 120.27%
fred_latest/fred_latest_curated:fred_latest_payrolls	147,314	+ 207.80%
fred_latest/fred_latest_curated:fred_latest_real_gdp	18,666.62	+ 498.69%

regime_fwd_returns

regime	count	avg_fwd_ret_...	median_fwd_ret_...	hit_rate_pct
Q2	1	-3.86%	-3.86%	0.00%

regime	count	avg_fwd_ret_...	median_fwd_ret_...	hit_rate_pct
Q3	1	+ 0.60%	+ 0.60%	+ 100.00%
Q4	1	+ 1.70%	+ 1.70%	+ 100.00%

surprise_index

timestamp	surprise_index
1994-01-01	2.54
1999-11-01	2.92
2000-04-01	0.27
2001-09-01	2.30
2003-07-01	2.07
2005-09-01	1.97
2008-07-01	0.15
2009-10-01	0.18
2011-07-01	-0.13
2013-12-01	1.92
2015-03-01	1.78
2015-08-01	1.62
2019-07-01	1.48
2020-01-01	1.51
2023-07-01	1.42

Statistics Summary

Key	Value
study	us_growth_nowcast_surprise
category	macro
status	success
parameters	{'lookback_years': 15, 'horizon': 21, 'growth_series': {'bea/bea_curated': ['gdp_real', 'exports_goods_services', 'personal_consumption_expenditures', 'private_fixed_investment_equipment'], 'bls/bls_curated': ['nonfarm_payrolls', 'avg_hourly_earnings_total_private'], 'fred_latest/fred_latest_curated': ['industrial_production', 'payrolls', 'real_gdp']}}
datasets	['bea/bea_curated', 'bls/bls_curated', 'fred_latest/fred_latest_curated']
rows	20751
git_commit	9178cf61eefd9b125a5087073e70ef1c9d91f687b
executed_at	2026-08-07T08:47:20.332956+00:00

Glossary

Term	Definition
n / count	Number of observations or events in the sample.
edge	Average excess return versus the unconditional baseline.
hit-rate	Share of events with a positive forward return.
p-value	Probability of observing the result under the null of no edge.
*** / ** / *	Significance at 99.9% / 99% / 95% confidence.
α (alpha)	Intercept or active return versus benchmark.
β (beta)	Sensitivity to a reference factor or market.
σ (sigma)	Volatility / standard deviation of returns.

Document Control

Field	Value
Report schema	2
Framework version	0.1.0
Classification	internal
Generated	2026-08-07 08:47 UTC
Author	Cerberus Research
Study slug	us_growth_nowcast_surprise
Run ID	20260807_094720
Git commit	978cf61e
Datasets	bea/bea_curated, bls/bls_curated, fred_latest/fred_latest_curated
Parameters hash	6f3e3522f011
Parameter · lookback_years	15
Parameter · horizon	21
Parameter · growth_series	{'bea/bea_curated': ['gdp_real', 'exports_goods_services', 'personal_consumption_expenditures', 'private_fixed_investment_equipment'], 'bls/bls_curated': ['nonfarm_payrolls', 'avg_hourly_earnings_total_private'], 'fred_latest/fred_latest_curated': ['industrial_production', 'payrolls', 'real_gdp']}
Data fingerprint	4cf6276cbaba
Duration	62.2s

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